# Tracking environmental processes in the coastal zone for understanding and predicting Oregon coho (*Oncorhynchus kisutch*) marine survival

E.A. LOGERWELL,[1,*] N. MANTUA,[2] P.W. LAWSON,[3] R.C. FRANCIS[1] AND V.N. AGOSTINI[1]

[1]*School of Aquatic and Fisheries Science, University of Washington, Seattle, Washington 98195, USA*
[2]*School of Marine Affairs/Joint Institute for the Study of the Atmosphere and Oceans Climate Impacts Group, University of Washington, Seattle, Washington 98195-4235, USA*
[3]*National Marine Fisheries Services, Northwest Fisheries Science Center, Newport, Oregon 97365, USA*

## ABSTRACT

To better understand and predict Oregon coho (*Oncorhynchus kisutch*) marine survival, we developed a conceptual model of processes occurring during four sequential periods: (1) winter climate prior to smolt migration from freshwater to ocean, (2) spring transition from winter downwelling to spring/summer upwelling, (3) the spring upwelling season and (4) winter ocean conditions near the end of the maturing coho's first year at sea. We then parameterized a General Additive Model (GAM) with Oregon Production Index (OPI) coho smolt-to-adult survival estimates from 1970 to 2001 and environmental data representing processes occurring during each period (presmolt winter SST, spring transition date, spring sea level, and post-smolt winter SST). The model explained a high and significant proportion of the variation in coho survival ($R^2 = 0.75$). The model forecast of 2002 adult survival rate ranged from 4 to 8%. Our forecast was higher than predictions based on the return of precocious males ('jacks'), and it won't be known until fall 2002 which forecast is most accurate. An advantage to our environmentally based predictive model is the potential for linkages with predictive climate models, which might allow for forecasts more than 1 year in advance. Relationships between the environmental variables in the GAM and others (such as the North Pacific Index and water column stratification) provided insight into the processes driving production in the Pacific Northwest coastal ocean. Thus, coho may be a bellwether for the coastal environment and models such as ours may apply to populations of other species in this habitat.

**Key words:** climate, coho salmon, General Additive Model, *Oncorhynchus kisutch*, Oregon, survival

## INTRODUCTION

In the past decade many insights into the complex dynamics linking variations in ocean conditions to variations in Pacific salmon (*Oncorhynchus spp.*) production have been gleaned from retrospective studies of climate and fishery data. Based on survival rates of hatchery and wild coho (*Oncorhynchus kisutch*) marked with coded wire tags (CWT), Coronado and Hilborn (1998) concluded that, not only were ocean conditions the dominant factor affecting coho survival in the 1970s and 1980s, but that survival variations were opposite between regions north and south of northern British Columbia. Hare *et al.* (1999) examined harvest data and showed that this pattern (a.k.a. 'inverse production regimes') tended to hold across all five species of commercially exploited NE Pacific salmon over most of the Twentieth Century, and that the swings between different production regimes are well correlated with swings between warm and cool phases of the Pacific Decadal Oscillation (PDO).

While these broad-scale studies have prompted renewed interest in ocean conditions in fishery research and management circles, it is our opinion that the utility of this information is limited due to a lack of application at the regional and local spatial scales common to fishery management activities. In the present study, our goal is the development of a simple environmental model that can be used to better understand local year-to-year changes in coho salmon

*Correspondence. e-mail: Libby.Logerwell@noaa.gov
Present address: E. A. Logerwell F/AKC2, Alaska Fisheries Science Center, PO Box 15700, Seattle, WA 98115-0070. USA.
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(*O. kisutch*) marine survival. To achieve this goal, we first constructed a conceptual model linking coho survival to environmental processes occurring during sequential time periods during early marine life history. We then quantified our conceptual model by building a General Additive Model (GAM) with specific environmental variables chosen *a priori* to represent processes occurring during each period. We use the results of the GAM-building process to gain insight into the environmental processes that have driven variability in coho survival in the past. We also apply current environmental data to our GAM to predict coho marine survival 1 year into the future.

### *Oregon Production Index (OPI) Coho salmon*

Coho salmon are a valuable resource economically, ecologically and culturally. In the Pacific Northwest (PNW) coho have provided extensive commercial and sport fisheries for most of the Twentieth Century, and have been an integral part of Native American culture for at least several thousand years. Widespread late-Twentieth Century declines in coho abundance in California, Oregon, Washington and British Columbia have prompted drastic restrictions in sport and commercial harvests, in some cases crippling the economies of fishing-dependent communities (National Research Council, 1996; Lichatowich, 1999; Taylor, 1999). Recently, serious concerns have been expressed about the long-term declines in abundance of a number of wild PNW coho stocks, warranting listings under the US Endangered Species Act (National Marine Fisheries Service, 2001).

Some of the most intensely studied coho populations in the world inhabit the OPI area, defined as the region from northern California to southern Washington (36.5–46.5°N) (Pearcy, 1992). The oceanography is dominated by the northern sector of the California Current, a highly productive coastal upwelling region downstream from the subarctic North Pacific and south of the West Wind Drift (Ware and MacFarlane, 1989). Oregon Production Index coho spawn in coastal streams and small tributaries of larger rivers during September–March. Migration of smolts downstream towards the sea occurs in late winter and spring, with a peak in May. Coho move rapidly through coastal estuaries into the sea where they mature over an 18-month period. Adults thus return to spawn during the fall following their second summer at sea.

### *Conceptual model*

To develop an environmental model to better understand variability in OPI coho survival, we formulated a scenario of four important periods during early marine life history:

1. Winter climate prior to smolt migration: Winter conditions have been hypothesized to influence water column stratification the following spring, which in turn influences ocean productivity (e.g. Polovina *et al.*, 1995; Gargett, 1997) and smolt habitat quality. We have termed this process 'winter preconditioning'.

2. Spring transition date: In the OPI area the spring transition between downwelling and upwelling favourable winds typically takes place in late March/early April, although the timing is highly variable (Bilbao, 1999). A mismatch in the timing of the onset of upwelling and smolt migration to sea could result in smolts not finding favourable marine habitat when they reach the ocean. Supporting this hypothesis are the results of Ryding and Skalski (1999) who related marine survival of CWT-marked coho from Washington hatcheries to time of the spring transition.

3. Ocean conditions during spring of smolt migration: Spring is regarded as a critical period during which coho mortality is thought to be particularly high and variable (Pearcy, 1992). Favourable conditions (strong upwelling winds, low water column stratification, low coastal sea level, strong equatorward transport and cool SST) are expected to correlate with high biological productivity and food availability for coho smolts. Previous studies have identified links between survival of PNW coho and coastal upwelling winds (Scarnecchia, 1981; Nickelson, 1986; Fisher and Pearcy, 1988; Holtby *et al.*, 1990), SST (Cole, 2000), and upper ocean stratification (Koslow *et al.*, 2002) during the spring of ocean entry.

4. Winter climate following smolt migration: Our hypothesis is that poor conditions during this overwintering period can result in poor survival to adulthood, either due to poor feeding conditions (Chelton *et al.*, 1982; Strub and James, 2000) or increased predation (Pearcy and Schoener, 1987; Pearcy, 1992, 1999). Previous studies (Lawson, 1997; Cole, 2000; Koslow *et al.*, 2002) showing that coho survival is related to SST and coastal sea level at the end of the maturing coho's first year at sea support this hypothesis.

The scenario described above is a synthesis of previous studies that identify important variables or processes during particular periods (e.g. spring transition or first winter at sea). We take the results of this large body of literature and combine them into one conceptual model spanning all seasons during the early marine life history of coho.

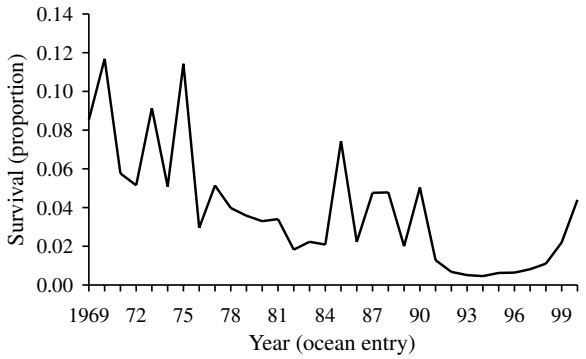
## DATA AND METHODS

Our work draws on smolt-to-adult survival estimates from the OPI. Although the OPI covers coho salmon entering the ocean from rivers in California, Oregon and southern Washington, since 1960 Columbia River hatcheries have produced the majority of smolts (Pearcy, 1992). Because most OPI coho are hatchery-reared, variability in smolt-to-adult survival due to changes in freshwater habitat has been virtually eliminated. Thus, variability in ocean habitat is expected to be the dominant driver of variability in OPI hatchery coho smolt-to-adult survival. We define the geographic extent of the ocean habitat as the Oregon/Washington coastal area. We thus assumed that juvenile and maturing coho occupied similar coastal areas. Pearcy and Fisher (1988) showed that most of the coho smolts entering the ocean from the Columbia River remain in waters off Oregon and Washington during their first few months at sea. Little is known about the migration patterns of maturing coho during their first winter at sea. However, information on ocean migrations later in the coho life cycle, after the first winter at sea (during the spawning migration) can be obtained from fishery recoveries of tagged coho. These data showed that the majority of Columbia River coho are distributed in Oregon and Washington waters (Weitkamp *et al.*, 1995).

### *Biological data*

Smolt-to-adult survival of OPI coho salmon was calculated from total hatchery releases within the OPI area and adult returns, adjusted for the wild salmon returns but not adjusted for returns of jacks in the previous year (< 10% of the adult return). The resulting time series for ocean entry years 1969–2000 shows a declining trend and periods of high interannual variation (Fig. 1).

**Figure 1.** Smolt-to-adult survival of Oregon Production Index coho salmon from ocean entry years 1969–2000 (corresponding to return years 1970–2001).




### *Oceanographic and atmospheric data*

We examined several oceanographic and atmospheric variables from which to select representatives for processes occurring during the four periods in our scenario: winter preconditioning, spring transition, spring upwelling season, and ocean conditions during the first winter at sea (see Table 1). These variables are described in more detail below. Fig. 2 shows the locations along the Washington–Oregon coast where each variable is measured.

### *North Pacific Index (NPI.JFM)*

The North Pacific Index (NPI) is the area-weighted sea level pressure over the region 30–65°N, 160°E–140°W (Trenberth and Hurrell, 1994). During winter months, the NPI is an index of the intensity of the Aleutian Low. Low values of the NPI indicate that the Aleutian Low is stronger. A strong Aleutian Low affects atmospheric circulation such that anomalously warm, moist air is advected over the NE Pacific and onto the west coast of the US, winter downwelling is strong, surface heat fluxes are weak, and upper ocean temperatures are anomalously high (Miller *et al.*, 1994). We calculated the mean of January–March NPI for the winters before and after smolt migration.

### *Upwelling wind indices (UW.JFM and UW.AMJ)*

We calculated mean upwelling winds for the period 1969 to present across three positions covering the Washington and Oregon coasts (42, 45 and 48°N; 125°W) and for two seasons, winter (January–March) and spring (April–June). Data are from http://www.pfeg.noaa.gov.

### *Sea surface temperature (SST.JFM and SST.AMJ)*

A time series of SST for the OPI area was derived from two sets of measurements at the mouth of South Slough, OR, located at 43°N (Tom Nickelson, Oregon Department of Fish and Wildlife, Corvalis, OR, pers. com.). Hourly sea temperatures for October 1993 to April 2000, measured at the US Coast Guard station (#9432780) at South Slough, were obtained from the NOAA Center for Operational Oceanographic Products and Services (http://co-ops.nos.noaa.gov). The water at the station is fully marine at high tide, so temperatures at daily high tides (also from the US Coast Guard station measurements) were selected and monthly averages calculated. To extrapolate the Coast Guard station time series back to 1969, a regression

**Table 1.** Short hand notations and brief definitions of all environmental variables examined. The extension 'JFM' indicates January–March means, and 'AMJ' indicates April–June means; 't0' indicates the year of ocean migration, and 't1' indicates the year after ocean migration.

| Variable name | Description |
|---|---|
| NPI.JFM.t0 | North Pacific Index, January–March mean, year of ocean migration |
| UW.JFM.t0 | Upwelling winds, 42–48° N mean, January–March mean, year of ocean migration |
| SST.JFM.t0 | Sea surface temperature, South Slough, Oregon, January–March mean, year of ocean migration |
| TRANS | Date of spring transition |
| UW.AMJ | Upwelling winds, 42–48° N mean, April–June mean |
| SST.AMJ | Sea surface temperature, South Slough, Oregon, April–June mean |
| SL.AMJ | Sea level at Neah Bay, Washington, mean residuals from April–June |
| NPI.JFM.t1 | North Pacific Index, January–March mean, year after ocean migration |
| UW.JFM.t1 | Upwelling winds, 42–48° N mean, January–March mean, year after ocean migration |
| SST.JFM.t1 | Sea surface temperature, South Slough, Oregon, January–March mean, year after ocean migration |

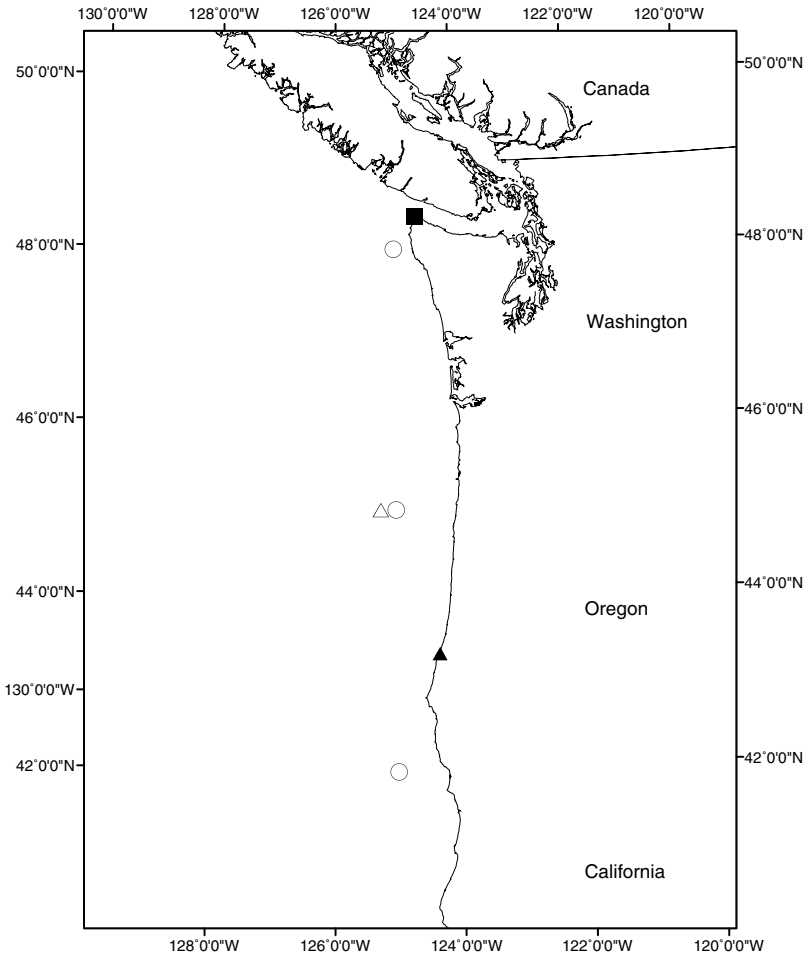



**Figure 2.** Washington–Oregon coast showing locations of environmental data sources. Solid square = Neah Bay sea level. Open circles = Upwelling indices. Open triangle = NCEP Pacific Ocean Analysis subsurface temperature and salinity. Solid triangle = South Slough station SST. Not shown is the location of the North Pacific Index (NPI) (30–65°N, 160°E–140°W).

equation was developed between the Coast Guard data and sea temperatures measured 0.8 nautical miles seaward of the Coast Guard station by the Oregon Institute of Marine Biology (OIMB) from 1969 to the mid-1990s. The regression equation, formulated for the period January 1994 to December 1997, is $y = 0.849x + 1.408$, where $y$ = Coast Guard SST and $x$ = OIMB SST, $R^2 = 0.84$.

To determine whether our time series of SST was representative of SST variability throughout the OPI region, we compared a subset of our data to a shorter time series of satellite-derived SST anomalies from 43 to 46°N and from the coast to 40 km offshore, 1985–99 (Cole, 2000; James Cole pers. com., 31 Penny St., Portsmouth PO1 2NH, UK). From the weekly anomalies of satellite SST, we calculated winter (January–March) and spring (April–June) mean anomalies across all latitudes. Regression of our point time series and the regional satellite series showed that the two data sets were comparable during both winter and spring. Regression $R^2$ for winter was 0.96 ($P < 0.0001$, d.f. = 13) and $R^2$ for spring was 0.71 ($P < 0.001$, d.f. = 13). The strong correlation between the winter averages of these two time series is consistent with the fact that variability in wintertime OPI coastal SSTs has a large regional coherence (Meuter *et al.*, 2002).

*Spring transition date (TRANS)*

The spring transition refers to a change from winter conditions dominated by high sea level, coastal downwelling and poleward winds to spring conditions characterized by low sea level, coastal upwelling and equatorward winds (Huyer *et al.*, 1979). To develop a time series of spring transition date for the OPI region, we looked for seasonal transitions in two time series: (1) area averaged daily upwelling indices from 42 to 48°N, 125°W (http://www.pfeg.noaa.gov), and (2) daily sea level residuals (corrected for the inverse barometer effect) measured at Neah Bay, WA (48°22.1′N 124°37.0′W; University of Hawaii Sea Level Center, http://uhslc.soest.hawaii.edu/). Examination of autocorrelation plots of daily upwelling winds and sea level indicated drops in autocorrelation at two different temporal lags, around 10 and 90 days, the latter likely reflecting seasonal variation. To filter out high frequency variation, a low pass filter with a stop frequency of 1/(10 days) was constructed (S-PLUS, MathSoft, Inc., Seattle, WA, USA). To draw out the seasonal pattern, a low pass filter with a stop frequency of 1/(90 days) was constructed. The date of spring transition was chosen as the date when the 1/(10 days) low pass filtered lines crossed zero. The 1/(90 days) low pass filter line confirmed that the selected date marked the beginning of a new seasonal state.

*Sea level (SL.AMJ)*

As an index of sea level variability resulting from alongshore winds/currents, Ekman transport, coastally trapped internal waves, and inflow to the California Current from the North Pacific current (Chelton and Davis, 1982; Simpson, 1984; Huyer and Smith, 1985), we calculated April–June means of daily sea level residuals at Neah Bay (see 'TRANS' above).

*Spring stratification*

In addition to spring upwelling winds, SST and sea level, we were interested in stratification because previous work (e.g. Gargett, 1997) suggested that winter climate conditions influence the strength of stratification the following spring which has implications for the biological effectiveness of wind-driven upwelling. However, a continuous time series of observed subsurface water properties does not exist for the coastal OPI region during 1969 and 2000 period of record. There is, however, a data set of modelled subsurface properties, the NCEP Pacific Ocean Analysis, with a temporal coverage of 1980–2000 (data provided by the NOAA_CIRES Climate Diagnostics Center, Boulder, Colorado, USA, http://www.cdc.noaa.gov/). We obtained temperature and salinity data from the NCEP grid point 45°N, 125.25°W and from these data calculated density profiles. These profiles showed that most of the seasonal variance in density occurred in the top 150 m and that densities averaged from the surface to 55 m well represented depth-averaged density down to 150 m. We calculated depth-averaged density anomalies over the upper 55 m by subtracting the long-term mean monthly depth-averaged density from each monthly depth-averaged density. A negative density anomaly represents relatively strong stratification, and a positive density anomaly represents weak stratification. We developed this proxy of ocean stratification from this shorter time series, not to include in our GAM, but to compare with our longer SST time series, and thus to investigate whether winter SST could be an indicator of spring ocean stratification.

*Model development: ocean entry years 1969–2000*

We used generalized additive modelling to develop a relatively simple 'ocean conditions' model for understanding and predicting variations in OPI coho marine survival. We chose to use this method because of the flexibility in defining predictor functions. Parametric and non-parametric terms, as well as linear and non-linear terms, can be combined into a single GAM (Chambers and Hastie, 1992). Statistical analyses were carried out in S-PLUS 2000 (MathSoft, Inc., Seattle, WA, USA).

To select a representative variable for each environmental process and to examine the shape of any non-linear effects, we examined bivariate scatterplots

of each of the environmental variables and coho survival. A linear least-squares regression, and a non-linear locally weighted least-squares regression, or loess, were fit to each. Of the suite of possible variables representing processes occurring during each of our four periods, we chose the one that had the highest $R^2$ (linear or non-linear). A parametric non-linear predictor function (b-spline) was chosen for variables that did not appear to be linearly related to coho survival because fewer degrees of freedom are used compared with loess. The degrees of freedom were chosen based on the shape of the loess fit in the original exploratory plot. The appropriateness of the non-linear function was confirmed by examining partial regression plots from the final GAM. If a straight line would not fit within the confidence limits of the (non-linear) partial regression for a particular variable then the non-linear function was retained.

After the predictor function for each environmental variable was chosen (linear or b-spline) we constructed a GAM of coho survival by sequential addition of each variable starting with winter preconditioning. Model fit and parsimony were evaluated with approximate *F*-tests and the Akaike information criterion (AIC). The *F*-test, conducted with each addition of a new variable, indicates whether the model is a significant improvement over the previous version (without the additional variable). The AIC statistic, also estimated with each addition, accounts for degrees of freedom used and the goodness of fit such that more parsimonious models have a lower AIC (Chambers and Hastie, 1992). Akaike information criterion is calculated in S-PLUS as $AIC = -2$ maximum by likelihood $+2$ number of parameters. Residual plots were examined for violations of model assumptions. Because the GAM was constructed with parametric functions, the fit of the final model was evaluated with ANOVA (*F*-test) and $R^2$ was calculated using S-PLUS commands for linear models (Chambers and Hastie, 1992).

*Forecasts: ocean entry year 2001*

To examine model predictions into the future, we applied environmental data from winter to spring 2001 to forecast ocean survival corresponding to 2002 adult returns. Because data for January–March 2002 coastal SST (winter after smolt migration) were not yet available, we made forecasts using the long-term mean coastal SST (1969–2001), mean + 1 SD and mean −1 SD. Forecast coho survivals (with pointwise standard errors) were estimated using what is known as 'safe' prediction in S-PLUS, which avoids incorrect predictions that could result from applying generic prediction methods to models which involve data-dependent transformations.

## RESULTS

A correlation matrix of all the environmental variables (Table 2) shows three sets of significantly intercorrelated variables representing three seasons: winter before smolt migration or 'winter preconditioning' (JFM.t0), spring (AMJ), and first winter at sea (JFM.t1). Correlations between indices at different seasons are not statistically significant, with the exception of winter (t0) and spring SST which are correlated somewhat weakly ($r = 0.39$). The correlation between TRANS and spring SST and sea level is partially an artifact of the way mean values of spring variables were calculated. Because the climatological mean spring transition date is March 31, an anomalously late spring transition means that the duration of the April–June upwelling season (with the associated low sea levels and cool SST) is truncated.

**Table 2.** Correlation matrix (Pearson's product moment correlation coefficient) for all environmental variables initially examined, 1969–2000. Shaded cells indicate groups of seasonal variables that are significantly correlated ($P < 0.05$).

| | NPI.JFM.t0 | UW.JFM.t0 | SST.JFM.t0 | TRANS | UW.AMJ | SST.AMJ | SL.AMJ | NPI.JFM.t1 | UW.JFM.t1 | SST.JFM.t1 |
|---|---|---|---|---|---|---|---|---|---|---|
| NPI.JFM.t0 | | 0.74 | −0.77 | −0.12 | 0.21 | −0.24 | −0.06 | −0.07 | −0.02 | 0.01 |
| UW.JFM.t0 | 0.74 | | −0.63 | −0.12 | 0.13 | −0.17 | −0.05 | −0.09 | −0.03 | −0.04 |
| SST.JFM.t0 | −0.77 | −0.63 | | 0.22 | −0.25 | 0.39 | 0.00 | 0.02 | −0.13 | 0.13 |
| TRANS | −0.12 | −0.12 | 0.22 | | −0.30 | 0.59 | 0.73 | 0.01 | 0.02 | 0.26 |
| UW.AMJ | 0.21 | 0.13 | −0.25 | −0.30 | | −0.56 | −0.56 | −0.15 | −0.11 | −0.18 |
| SST.AMJ | −0.24 | −0.17 | 0.39 | 0.59 | −0.56 | | 0.50 | 0.09 | 0.07 | 0.22 |
| SL.AMJ | −0.06 | −0.05 | 0.00 | 0.73 | −0.56 | 0.50 | | −0.06 | −0.04 | 0.26 |
| NPI.JFM.t1 | −0.07 | −0.09 | 0.02 | 0.01 | −0.15 | 0.09 | −0.06 | | 0.74 | −0.76 |
| UW.JFM.t1 | −0.02 | −0.03 | −0.13 | 0.02 | −0.11 | 0.07 | −0.04 | 0.74 | | −0.63 |
| SST.JFM.t1 | 0.01 | −0.04 | 0.13 | 0.26 | −0.18 | 0.22 | 0.26 | −0.76 | −0.63 | |

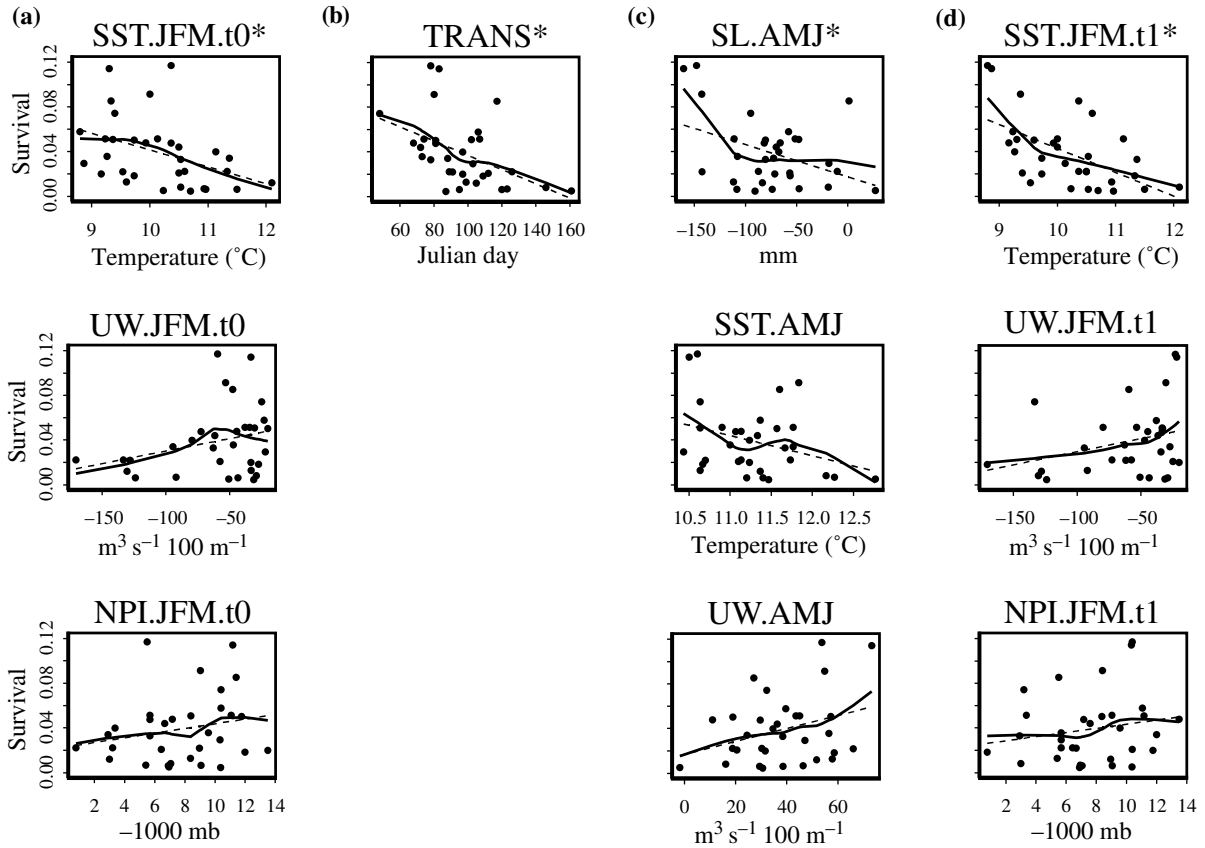
*Exploratory data analysis*

To select a representative variable for environmental processes occurring during each of the four periods in our conceptual model, we examined bivariate scatterplots of each of the environmental variables and coho survival (Fig. 3). All three variables representing winter preconditioning appeared to have linear relationships with coho survival (Fig. 3a). Weaker Aleutian Low (greater NPI values), weaker downwelling (less negative NCEP values), and colder SST were associated with higher coho survival. The linear regression of coho survival and winter SST was statistically significant ($R^2 = 0.17$, $F = 6.07$, d.f. $= 1$ and 30, $P < 0.05$). The regressions between coho survival and NPI and upwelling indices were not. Thus, we chose SST to represent winter preconditioning in the GAM. There was a statistically significant, linear relationship between spring transition date and coho survival ($R^2 = 0.23$, $F = 9.10$, d.f. $= 1$ and 30, $P < 0.01$), later dates were associated with lower survival (Fig. 3b).

Of the three variables representing the spring upwelling season, sea level was the only one that appeared to have a strongly non-linear relationship with coho survival (Fig. 3c). The loess regression was statistically significant with multiple $R^2 = 0.39$ (Npar $F = 5.08$, $P < 0.05$). There were no significant relationships (linear or non-linear) between coho survival and the other spring variables, SST and upwelling indices (Fig. 3c). Thus, sea level was chosen to represent spring coastal ocean conditions. A b-spline with one knot at −95 mm and degree = 1 (d.f. = 3) was chosen as the most parsimonious non-linear smoother.

There was a statistically significant linear relationship between coho survival and SST during the first winter at sea ($R^2 = 0.33$,$F = 14.7$, d.f. $= 1$ and 30, $P < 0.01$). There was no significant relationship between coho survival and the other two variables representing t1 winter conditions (NPI and upwelling indices, Fig. 3d). Thus, SST was chosen to represent post-smolt winter conditions in the GAM.

**Figure 3.** Bivariate relationships between environmental variables and coho survival. Solid lines are loess smoothers (span = 0.75, degree = 1) and the dashed lines are linear fits. Asterisk indicate relationships (linear or loess) that are statistically significant at $P < 0.05$. Variables represent four environmental processes: (a) winter preconditioning (January–March before smolt migration), (b) spring transition, (c) spring (April–June), (d) first winter at sea (January–March after smolt migration).



*Building the General Additive Model*

The GAM was assembled starting with presmolt migration winter SST (Table 3). With each addition of the variables chosen to represent subsequent periods, residual deviance decreased, and the AIC statistic decreased, the two statistics indicating increasingly better model fit and parsimony. The shapes of the relationships between coho survival and the environmental variables retained in the GAM are shown in partial regression plots (Fig. 4). Each plot is the contribution of a term to the additive predictor, and the *y*-axis label shows the expression used to specify the predictor in the model formula. A comparison of Figs 4 and 3 shows the similarity between the parametric b-spline and the non-parametric loess smoother examined during exploratory data analysis.

The final GAM, shown on the last line of Table 3, resulted in a model that explains a high and statistically significant proportion of the variance. Recall that the test statistics shown in Table 3 measure the improvement of model fit with each additional parameter, not the statistical significance of the final model $R^2$. The final model multiple $R^2$ was 0.75 ($F = 15.75$,$P < 0.0001$, d.f. = 5 and 26). A comparison of observed versus predicted OPI coho survival time series shows that the GAM explained the overall decline in coho survival from 1969 to 2000 and much of the interannual variation (Fig. 5). One notable error is the GAM prediction for 1999, which is substantially higher than the observed survival.

Residual plots do not show strong violations of model assumptions (Fig. 6a,b). A plot of autocorrelation versus time lag in years (Fig. 6c) shows that any autocorrelation in the original time series of coho survival was removed by the GAM, suggesting that variability due to the environment was correctly modelled (Bakun, 1996).

*Forecast results: ocean entry year 2001*

Our forecast for ocean entry year 2001 (2002 adult returns) was 6–8% when the mean (1969–2000) JFM.SST.t1 was used to predict ocean survival. Survival was higher, 7–9%, when the mean −1SD was used; it was lower 4.5–8.5% when the +1SD was used.

## DISCUSSION

Our GAM model using ocean conditions explains 75% of the variability in OPI coho survival from 1969 to 2000. The parameters, in seasonal sequence, are: (1) January–March SST during the winter before smolt migration to sea, (2) the date of the spring transition, (3) April–June sea level during the spring of smolt migration, and (4) January–March SST during the winter after smolt migration. The model captures both the downward trend in OPI coho survival since the mid-1970s and the patterns of interannual variation throughout the period of study.

Observed survivals fall within the 95% confidence limits of our GAM in almost all years, with the notable exception of ocean entry year 1999. Our prediction for survival of 1999 smolts (returning as adults in 2000) was substantially higher than the observed survival. One possible explanation is that zooplankton numbers were not particularly high in the summer of 1999, despite apparently favourable ocean conditions (William Peterson, Northwest Fisheries Science Center, National Marine Fisheries Service, Newport, Oregon, pers. com). Peterson suggests that the subarctic zooplankton community was not able to immediately 'rebound' from very low abundances resulting from several years of warm water conditions previous to 1999.

Our results build on a substantial literature describing the processes affecting survival of PNW coho during various stages of early marine life. For instance, early studies identified a link between coastal upwelling winds and survival of PNW coho for the 1960–81 period of record (Scarnecchia, 1981; Nickelson, 1986; Fisher and Pearcy, 1988; Holtby *et al.*, 1990). However, the relationship between upwelling wind indices and OPI coho survival has not held since the mid-1980s, so it is clearly not possible to explain OPI coho survival in terms of coastal wind forcing alone

**Table 3.** Construction of the GAM. As each term is added, deviance, AIC and the F-statistic are examined to determine whether the term will be included in the final model. Response: OPI coho survival.

| Terms | Residual d.f. | Residual deviance | AIC | *F* | *P* |
|---|---|---|---|---|---|
| SST.JFM.t0 | 30 | 0.024 | 0.028 | 6.07 | 0.020 |
| SST.JFM.t0 + TRANS | 29 | 0.019 | 0.024 | 7.29 | 0.011 |
| SST.JFM.t0 + TRANS + SL.AMJ | 27 | 0.009 | 0.016 | 14.24 | <0.001 |
| SST.JFM.t0 + TRANS + SL.AMJ + SST.JFM.t1 | 26 | 0.007 | 0.011 | 7.86 | 0.009 |

(Lawson, 1997). Cole (2000) analysed the relationship between coho survival and satellite-derived nearshore SST over a relatively short period (1985–96). The results were that cold SST during the fishes' first spring in the ocean was associated with high survival and that warm SST from January to June of their second year at sea was apparently detrimental to survival. However, a hindcast of survival before 1985 based on Cole's (2000) temperature model did not match observations and a forecast did not agree with predictions based on jack returns, indicating that SST alone can not fully explain variability in coho survival (Cole *et al.*, 2001).

Koslow *et al.* (2002) have developed a multivariate method for linking variations in coho marine survival to the ocean environment. Instead of focusing on a single environmental variable, the authors synthesize 42 environmental time series from central California to Washington in a Principal Components Analysis (PCA). They showed significant linear relationships between coho survival and PC's that represent a linear combination of coherent variations in: (1) upper ocean stratification, SST and upwelling favourable winds in the spring of ocean entry, and (2) SST and coastal sea level in the winter of the first year at sea. Thus, our results are consistent with Koslow *et al.* (2002), and we expand the seasonal scope by incorporating processes occurring during two additional periods (winter preconditioning and spring transition).

There are also some important differences in scale and methodology between our work and Koslow *et al.* (2002). Because the majority of coho smolts released from OPI hatcheries enter the ocean from the Columbia River, we restricted our environmental time series to the latitudes of Washington and Oregon and thus our analysis is at a relatively local scale. Koslow *et al.* 's analysis was at a larger, regional scale – their time series spanned nearly the entire US west coast. In fact, their first PC for spring data was dominated by physical variability in the southern half of the data range (Cape Blanco to Pt. Conception). The timing and magnitude of seasonal cycles of temperature,

**Figure 4.** Partial regression plots for the GAM of coho survival illustrated in Table 3. 95% confidence limits (dashed lines) and partial residuals around the fitted lines are shown. The *y* -axis label is the expression used to specify each variable in the GAM formula. bs() indicates a 1-degree b-spline smoother with the specified knots. SST.JFM.t0, TRANS, and SST.JFM.t1 were fit using a linear term.

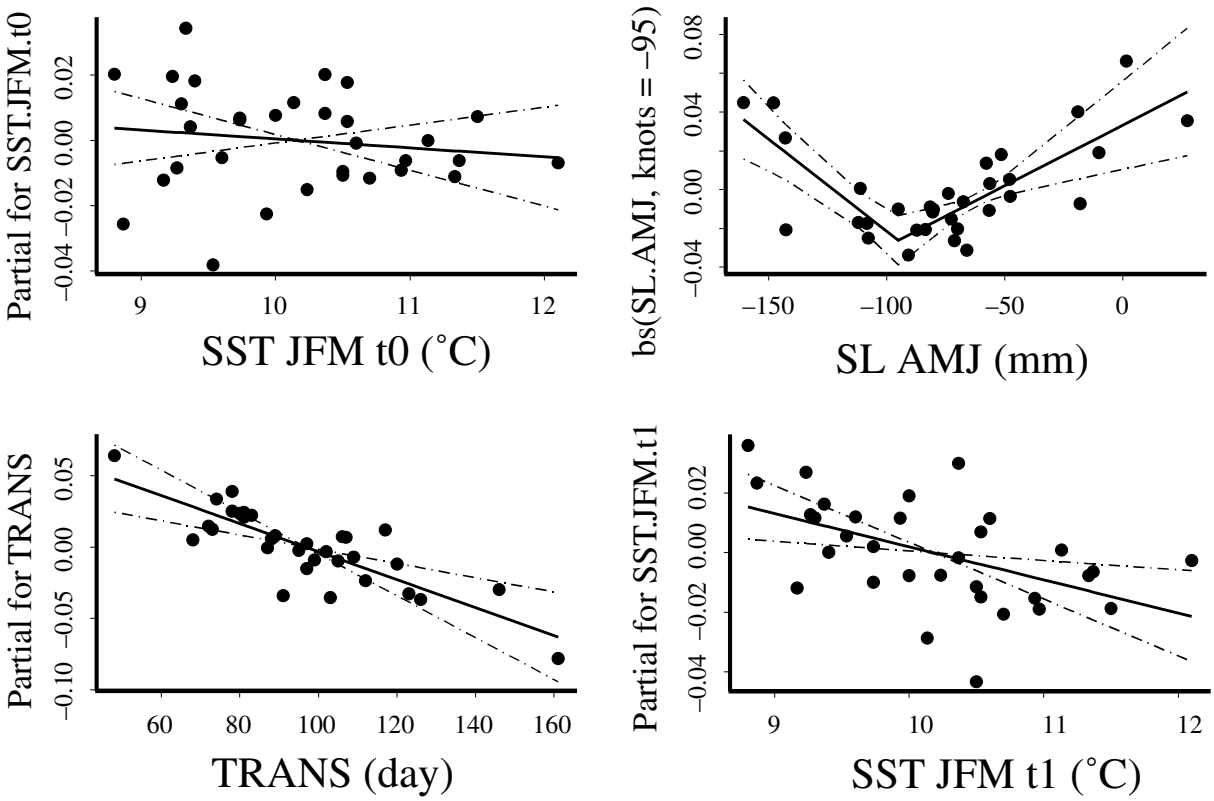
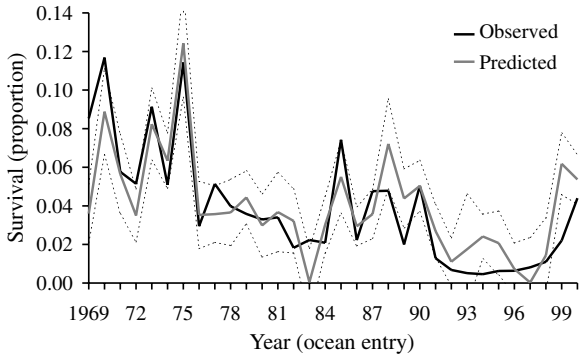


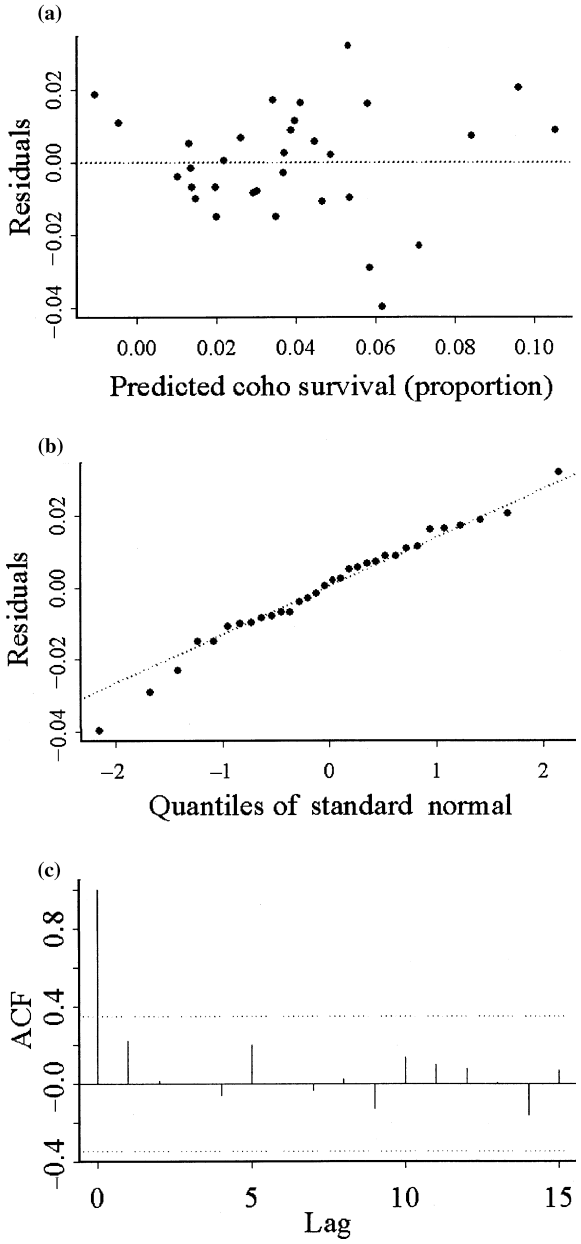
**Figure 5.** Time series of observed OPI coho survival compared with survival predicted by the GAM model. Ninety-five per cent confidence limits around predictions are indicated by the fine dotted line.




winds and sea level vary substantially with latitude from Washington to California (Strub *et al.*, 1987) such that fluctuations in these variables are not correlated between sites in California and Washington or Oregon (Osmer and Huyer, 1978). Meuter *et al.* (2002) similarly showed that spring/summer coastal physical variables from Washington to Alaska (upwelling, SST and salinity) were correlated at spatial scales <1000 km, but were not correlated at larger scales. Correlation patterns for survival rates of salmon also showed poor correlation at scales of 1000 km and greater. Thus, we felt that it was important to build a model of coho survival with locally measured variables. This potential scale mismatch might partially explain the lower explained variance in OPI survival achieved by the Koslow *et al* regression ($R^2 = 0.47$) compared with our GAM ($R^2 = 0.75$).

The other important difference between our work and that of Koslow *et al.* (2002) is in the conceptual and quantitative methods that were used to arrive at the results. Before building a quantitative model, we formulated a conceptual and mechanistic model describing a temporal series of environmental processes during early life history that determine coho survival. We then selected, *a priori*, the relevant environmental time series for each process. In this way, the conceptual model served as a hypothesis that we tested with a quantitative model (GAM). In contrast, Koslow *et al.* (2002) used the results of PCA of a large number of environmental time series to create and select PCs that represent coherent variability among groups of environmental variables. Their approach was to synthesize a large array of environmental data, identify patterns of variability and then examine whether coho survival is related to those patterns.

**Figure 6.** Diagnostic plots for the GAM. (a) residuals plotted against predicted coho survival, (b) quantiles of the standard normal distribution, (c) first-order autocorrelation functions (ACF) of the residuals plotted against time lag in years, dotted lines indicate statistically significant levels of autocorrelation.




An advantage of our approach is that our GAM can produce predictions of future coho survival from three environmental time series and is thus a relatively simple forecasting tool. Our forecast of survival for adult return year 2002 (2001 ocean entry year) ranged from 4.5 to 9% over a range of potential 2002

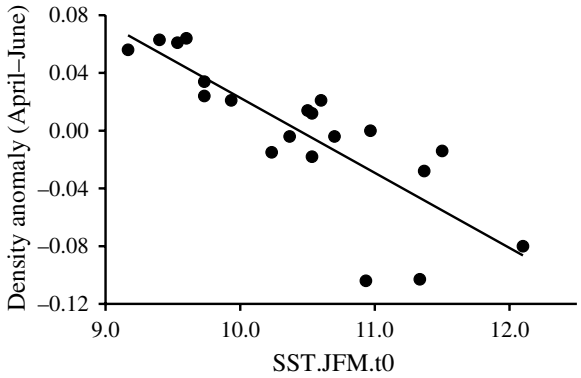
winter SST (long-term mean ± 1SD). This forecast is considerably higher than observed coho survival in the early 1990s and within the range observed during the early 1970s and late 1980s, suggesting continuing recovery of OPI coho. However, our forecast is substantially higher than that predicted by the return of precocious males ('jacks'), fish that entered the ocean as smolts in spring 2001 and returned in fall of the same year (1 year earlier than adult fish). The jack-based prediction for adult survival for return year 2002 is 1.4% (Pacific Fishery Management Council, 2002). It will not be known until fall 2002 which prediction is most accurate. However, adult survival predictions based on jack returns have proven skilful in the past (Pacific Fishery Management Council, 2002). One advantage of an environmentally based predictive model such as our GAM is the potential for linkages with climate models that could result in forecasts more than 1 year in advance of adult returns. Predictions of these sorts, accurate even to an order-of-magnitude, would provide valuable support for salmon harvest management and restoration strategies.

Implicit in our analysis is the assumption that ocean parameters such as SST and sea level are correlated with changes in ocean productivity and coho prey. Previous studies that demonstrated a link between climate variability and the abundance of central North Pacific phytoplankton (Polovina *et al.*, 1995), subarctic zooplankton (Brodeur and Ware, 1992), and British Columbia zooplankton (Mackas *et al.*, 2001) support this assumption. Peterson and Mackas (2001) found that the species composition of the zooplankton community off Oregon and Vancouver Island varied with changing ocean conditions. In the 1970s and 1980s, subarctic zooplankton communities – including euphausiids that are known to be important prey species for maturing coho (*e.g. Thysanoessa spinifera* and *Euphausia pacifica*) – dominated during the spring/summer upwelling season. During the extremely low coho survival/warm coastal SST era of 1992–98, the spring–summer abundance of subarctic species was anomalously low, while subtropical species became increasingly common in the coastal zone. This situation appears to have reversed itself in late 1998 and early 1999 – the dominance of boreal species in the spring/summer has been the rule through the summers of 2000–01 (William Peterson, Alaska Fisheries Science Center, National Marine Fisheries Service, Newport, OR, pers. comm.). Also implicit in our analysis is that survival of OPI coho smolts is not density-dependent. Pearcy (1992) reviewed the evidence and concluded that although the issue is still being debated, there is not strong support for density-dependence of OPI coho survival.

In addition to providing a method for forecasting coho survival, our results provide insight into the mechanisms that have driven variability in coho survival in the past. We discuss these in terms of our four environmental processes:

1. Winter climate prior to smolt migration: Our study is the first to examine the impacts of presmolt ocean climate on coho marine survival. Our model indicates that cooler SST during the winter prior to smolt migration contributes to higher survival. This environmental process was included in our scenario of coho survival because winter conditions have been hypothesized to influence water column stability the following spring (Polovina *et al.*, 1995; Gargett, 1997). This is the process we have termed 'winter preconditioning'. To test this hypothesis we compared the mean April–June 0–55 m density anomaly from the NCEP Pacific Ocean Analysis (1980–2000) to the previous winter SST (SST.JFM.t0) (see Fig. 7). There was a strong and statistically significant relationship between the two ($R^2 = 0.67$, $P < 0.001$, $N = 20$), with warmer winter SST associated with greater spring water column stratification (lower than average density in the upper 55 m). To our knowledge, we provide the first empirical evidence for a strong link between winter ocean conditions and stratification the following spring. The following mechanism for this link is suggested by the correlations between ocean and atmospheric variables: relatively weak downwelling (poleward) winds in the winter, due to a weaker Aleutian Low, result in less onshore transport of warm oceanic water, and reduced upper ocean stratification

**Figure 7.** Sea surface temperature during the winter before smolt migration (SST.JFM.t0) versus upper water column density anomaly the following spring. Line fitted by least-squares regression.



the following spring. We caution that our measure of stratification is primarily supported by output from an ocean simulation model, and although the model is informed by subsurface observations, a detailed collection and analysis of subsurface water column structure is likely needed to better understand the process of winter preconditioning.

2. Spring transition date: A later transition between winter downwelling and spring upwelling season appears to contribute to relatively poor coho survival, perhaps due to a temporal mismatch between arrival of smolts to sea and seasonal increases in ocean food production in the nearshore upwelling habitat.

3. Ocean conditions during spring of smolt migration: Our model indicates that lower spring sea level anomalies, below a threshold of around −100 mm, are correlated with higher coho survival. Low sea level can result from strong southward winds/currents in the alongshore direction, strong offshore Ekman transport, and increased inflow to the California Current from the North Pacific current. These processes are expected to result in enhanced feeding conditions due to the transport of nutrients from depth to the euphotic zone, and/or the equatorward transport of nutrients and boreal copepod species from subarctic waters (Chelton *et al.*, 1982; Chelton and Davis, 1982; Simpson, 1984; Huyer and Smith, 1985). Because reduced sea level can be the result of both increased upwelling and southerly transport, it is difficult to distinguish the relative impacts of variations in wind-driven upwelling and variations in the transport of the California Current. Analysis of variability in water mass tracers such as salinity might help resolve this issue.

4. Winter climate following smolt migration: Our results suggest that cooler SST during the winter following smolt migration contributes to high coho survival. The mechanisms may involve either improved feeding conditions or reduced predation.

**Table 4.** Environmental parameters from the GAM and OPI coho survival. +/− refers to above/below the median value for linear parameters (SST.JFM.t0, TRANS, SST.JFM.t1) and above/below the breakpoint or knot for non-linear parameters (SL.AMJ). Thus, '−' indicates cooler SST, earlier spring transition, and lower sea level. Shaded cells mark favourable states of each variable. The rows of the table are ordered by decreasing OPI coho survival.

| Year | SST.JFM.t0 | TRANS | SL.AMJ | SST.JFM.t1 | OPI Marine | Survival |
|---|---|---|---|---|---|---|
| 1970 | + | − | − | − | 0.117 | + |
| 1975 | − | − | − | − | 0.114 | + |
| 1973 | − | − | − | − | 0.091 | + |
| 1969 | − | + | + | + | 0.085 | + |
| 1985 | − | − | − | + | 0.074 | + |
| 1971 | − | + | + | − | 0.058 | + |
| 1972 | − | + | + | − | 0.051 | + |
| 1977 | − | − | − | + | 0.051 | + |
| 1974 | − | + | + | − | 0.051 | + |
| 1990 | − | − | + | − | 0.050 | + |
| 1988 | − | − | + | − | 0.048 | + |
| 1987 | + | − | + | − | 0.048 | + |
| 2000 | + | − | + | − | 0.044 | + |
| 1978 | + | + | + | − | 0.040 | + |
| 1979 | − | − | − | + | 0.036 | + |
| 1981 | + | − | + | − | 0.034 | + |
| 1980 | + | − | + | + | 0.033 | − |
| 1976 | − | + | + | − | 0.029 | − |
| 1983 | + | + | + | + | 0.022 | − |
| 1986 | + | − | + | + | 0.022 | − |
| 1999 | − | − | − | + | 0.022 | − |
| 1984 | + | + | + | − | 0.021 | − |
| 1989 | − | + | + | − | 0.020 | − |
| 1982 | − | + | + | + | 0.018 | − |
| 1991 | − | + | − | + | 0.013 | − |
| 1998 | + | + | + | − | 0.012 | − |
| 1997 | + | + | + | + | 0.008 | − |
| 1992 | + | + | + | + | 0.007 | − |
| 1996 | + | + | + | + | 0.006 | − |
| 1995 | + | − | − | + | 0.006 | − |
| 1993 | + | + | + | + | 0.005 | − |
| 1994 | + | − | + | + | 0.005 | − |

Feeding conditions could vary with the transport and nearshore entrainment of zooplankton-rich subarctic water from the north (Chelton *et al.*, 1982; Strub and James, 2000). Predation could increase with the invasion of the coastal zone by offshore and/or subtropical pelagic predator species during years of exceptionally warm SST (Pearcy, 1992, 1999; Pearcy and Schoener, 1987).

One point that deserves further discussion is that the variables representing each seasonal process (presmolt winter SST, spring transition date, spring sea level, and post-smolt winter SST) were essentially uncorrelated with each other. This distinguishes our work from previous studies that model the effects of intercorrelated variables on coho survival (e.g. Ryding and Skalski, 1999; Koslow *et al.*, 2002). To further illustrate this point, we have categorized each process as in a state favourable to coho survival (according to our model) or unfavourable, where favourable is above the time series median and unfavourable is below (Table 4). Similarly, coho survival is categorized as 'good' or 'poor'. In 2 years, all conditions were favourable, resulting in some of the highest survival rates of the time series (1975 and 1973). In other years, all conditions were unfavourable, resulting in some of the lowest survivals (1983, 1997, 1992, 1996, 1993, in order of decreasing survival). Nonetheless, there are many years when only some conditions were favourable/unfavourable and survival was good/poor. One striking pattern is that SST during winter before smolt migration was favourable (cool) in almost all years when survival was around 5% or greater. However, this variable alone does not tell the whole story, as some low survival years (1976, 1999, 1989, 1982, 1991) had favourable winter SST.

A substantial body of literature shows that there are decadal-scale regimes of salmon production in the NE Pacific, regimes which can be explained by decadal scale climate variability and coherent changes in ocean processes (Beamish and Bouillon, 1993; Francis and Hare, 1994; Hare and Francis, 1995; Gargett, 1997; Mantua *et al.*, 1997; Hare *et al.*, 1999; Beamish *et al.*, 1999; Beamish *et al.*, 2000). Our results are consistent with the regime hypothesis, but we also show that oceanographic processes can explain year-to-year variations in salmon survival. Coho survival appears to be impacted by processes occurring during a series of time periods from the winter before the smolts migrate to sea through spring and into their first winter at sea. Conditions during all of these periods are not necessarily required to be good to result in good survival or poor to result in poor survival. Our results suggest specific mechanisms that contribute to coho survival variability, such as winter preconditioning or the timing of the onset of the upwelling season. Process-oriented field studies are needed to establish whether and how these mechanisms function – whether there are direct links between oceanography, marine habitat, food availability, and survival or whether predator dynamics interact with oceanographic processes. The results of our study and many of the studies that precede ours make one thing clear: the survival and production of PNW coho is strongly related to ocean conditions. Because of this coho can serve as a bellwether species for variability in the coastal physical and biological environment.

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